

Internship Roadmap

Intern Name – Srushti Dandekar, Kavya Durga Sri Gullapalli

Project Title – City-Scale Retrofit Analysis of Ahmedabad

Software and External Resources:

1. Study Materials and Reference papers:

- a. [City-Scale Building Retrofit Analysis: A Case Study Using CityBES](#)
 - b. [Assessing a fit-for-purpose urban building energy modelling framework with reference to Ahmedabad.pdf](#)
 - c. Design book of retrofit building guidelines.
2. QGIS Software
 3. Python Libraries – GeoPandas, Pandas, NumPy, Matplotlib
 4. EnergyPlus Simulation Software
 5. Google Earth Pro
 6. OpenStreetMap (OSM)
 7. Ahmedabad EPW Weather File data

Objectives –

Week 1 & Week 2 –

1. Develop a basic understanding of Urban Building Energy Modeling (UBEM) and retrofit analysis.
2. Study CityBES workflow and Ahmedabad climate conditions.
3. Understand GIS-based city-scale mapping & energy analysis techniques.

Deliverables – Foundational learning summary on UBEM, GIS and retrofit analysis techniques.

Week 3 & Week 4 –

4. Collect Ahmedabad residential building footprints from OpenStreetMap.
5. Download satellite imagery and building geometry data.
6. Collect Ahmedabad EPW weather data and analyze climatic parameters.
7. Create an initial GIS database for Ahmedabad residential buildings.

Deliverables – Ahmedabad residential building GIS database and weather data collection summary.

Week 5 & Week 6 –

8. Prepare GIS datasets and validate building geometries.
9. Develop baseline residential building energy models in CityBES.
10. Generate baseline energy consumption reports.
11. Develop Python program script for CityBES

Deliverables – Baseline building energy model report and energy performance analysis.

Week 7 & Week 8 –

12. Compile final technical report and simulation observations.
13. Deliver final presentation on Ahmedabad city-scale retrofit analysis.
14. Compare baseline and retrofit energy performance.
15. Analyze cooling load reduction and carbon emission savings.

Deliverables – Retrofit simulation results and comparative energy savings analysis. Final technical report, GIS heat maps, Final documentation and project presentation.

Presentation 1

Mid-term presentation covering baseline energy modeling, GIS database preparation and simulation workflow. [week-4].

Presentation 2

Final presentation covering retrofit simulations, energy savings, carbon reduction and future recommendations. [week-8].